

Interactive Digital Art and Design SE608

Project Proposal Year IV



Student Name:	Riona Kelly	Student Number:	C00296706
Working Title:	Ri-Connect!		
Description:	This will be a physical device functioning as a Virtual Pet similar to a 'Tamagotchi'. However, it differs by featuring a USB Connector that can be plugged in to a computer where your pet may be interacted with in new ways. It will feature nostalgic retro aesthetics with unique creatures that the players may receive as their pets. The physical device will feature a microcontroller with a 1.47 inch LCD screen and USB connector, three buttons, a vibration motor, and a battery, all fit into pocket-sized 3D printed case modelled in (most likely) SolidWorks. I plan for this to be something that you carry around during the day, taking care of your pet while they keep you company unobtrusively, only vibrating for attention with the option to turn this off. Additionally it will charge when connected to a computer, necessitating this physical connection and encouraging interacting with both ways to play with this device. When connected to a computer through the USB slot, the user may open a website (maybe an app but I'm planning website for now so it'll work on any computer) that will recognise that the user has a Ri-Connect device plugged in and "transfer the pet from the device to your computer" where the user may play, take care of, and interact with their pet in more ways with more depth. The code for the microcontroller will be developed with Arduino. I would like to implement as many features as I can into the device but because of its RAM, I may be limited in some aspects and need to spend a lot of time optimising and leave a lot of the more intense features for the website counterpart.		
Reasons for selecting project:	I selected this to be my final year project because I have always found the subject of virtual pets and people's response to them extremely interesting and loved creating my virtual pet for the last project. It is a genre that I am incredibly attached to and even though I love the virtual pets I have I always feel like there are parts that I don't like and would want to change and improve, and with what I am studying I think it would be a shame to not try my best to do just that. I also really like the idea of making a tangible device that people can interact with. I have only ever created digital programs for		

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	projects like this before and just the idea of having a physical item that people can hold and see what I have created sounds so fulfilling and interesting, and I welcome the challenge. I had this idea over summer and have not been able to stop thinking about it since, and I know I will not be able to stop thinking about it until it is brought to fruition. I feel like I have enough experience and drive to complete this fully even though I will need to learn new skills while doing it.
Proposed research content:	I want to study the success of old and new virtual pets, learn how they became such a cult phenomenon, and implement these findings into my own project to create a nostalgic toy with an engaging modern twist. (See articles below)
External links (if applicable):	1. Allison, A. (2006) <i>Millennial Monsters: Japanese Toys and the Global Imagination</i> by Anne Allison, University of California Press. doi: 10.1111/J.1548-1425.2008.00110.X. 2. Dodd, M., Fowler, A. and Lottridge, D. (2025). <i>Purr-ogrammed love: A narrative review of virtual pets</i>, [online] Science Direct. https://www.sciencedirect.com/science/article/pii/S1875952125000382, doi:10.1016/j.entcom.2025.100958 [Accessed 20 Sept. 2026].. 3. Kusahara, M. (2001). <i>The art of creating subjective reality: An analysis of Japanese digital pets</i>, [online] MIT. https://direct.mit.edu/leon/article-abstract/34/4/299/44128/The-Art-of-Creating-Subjective-Reality-An-Analysis, doi:10.1162/00240940152549203 [Accessed 20 Sept. 2026]. 4. Ocheretyany, K.A. (2024). <i>Ghost in my pocket: Tamagotchi media archaeology</i>, [online] Galactic Media. https://galacticamedia.com/index.php/gmd/article/view/492, doi:10.46539/gmd.v6i2.492 [Accessed 20 Sept. 2026]. 5. Pettman, D. (2009). <i>Love in the Time of Tamagotchi</i>, [online] Academia.edu. Available at: https://www.academia.edu/2344255/Love_in_the_Time_of_Tamagotchi. doi:10.1177/0263276409103117 [Accessed 20 Sept. 2026]. 6. Ponzel, G. (2025). <i>Love on batteries — The story of</i>

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	<i>tamagotchi and digital attachment. Issue of magazine №19 global technology. [online] The Global Technology. Available at: https://theglobal.technology/issue-19/technologies/tamagotchi-and-love-on-batteries/ [Accessed 20 Sept. 2026].</i> 7. Reider, N.T. (2026). <i>Animating Objects: Tsukumogami ki and the Medieval Illustration of Shingon Truth</i>. [online] Nanzan-u.ac.jp. Available at: https://nirc.nanzan-u.ac.jp/journal/6/issue/179/article/1278 [Accessed 20 Sept. 2026]. 8. Schmauks, D. (2000). <i>Teddy bears, Tamagotchis, transgenic mice: A semiotic typology of artificial animals</i>. <i>Sign Systems Studies</i>. [online]. Available at https://ojs.utlib.ee/index.php/sss/article/view/SSS.2000.28.17, doi:10.12697/sss.2000.28.17 [Accessed 20 Sept. 2026]. 9. Ying, H., Feng, X., Lou, Q., Cai, L. and Zhang, X. (2024). <i>Arpet: Using Virtual Pets to Facilitate F2F Social Communication in PAR</i>. [online] IEEEExplore. Available at: https://ieeexplore.ieee.org/document/11027832 [Accessed 20 Sept. 2026].
Hardware requirements:	• ESP32-S3-LCD-1.47 Microcontroller (owned) • 3D Printer + Filament • Soldering Equipment • Three Simple Buttons • Vibration Motor • Battery • Wiring • MicroSD Card
Software requirements:	• Aseprite • Arduino IDE • Visual Studio • SolidWorks • OrcaSlicer
Other requirements:	

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Signed: Riona Kelly

Date: 20/09/2026

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Approved/Not Approved:		
Reasons for not approving project:		
Conditions attached to approving project:		
Approved/Not Approved:		
Name of Supervisor:		
Signed: _____		Date: _____